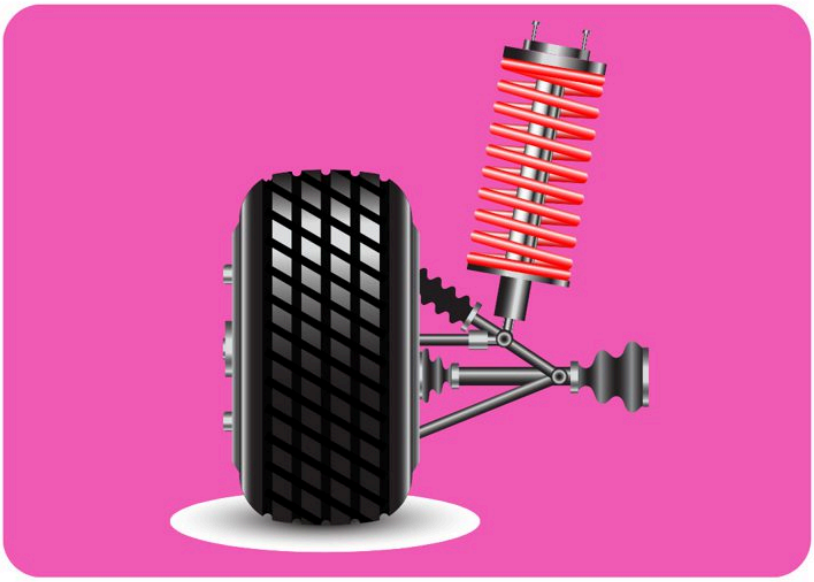
Chapter 18

Oscillations



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A Level Physics
Prepared and taught by James Gan

Chapter Outline

18.1 Types of Oscillations
18.2 Oscillations in Action
18.3 Characteristics of Oscillatory Motion
18.4 Fundamentals of Simple Harmonic Motion (SHM)
18.5 Visualising Simple Harmonic Motion
18.6 Angular Frequency
18.7 More Equations
18.8 Conversion of energy in SHM
18.9 Damped Oscillations
18.10 Resonance

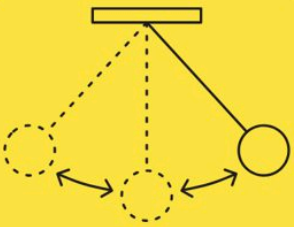


18.1 Types of Oscillations

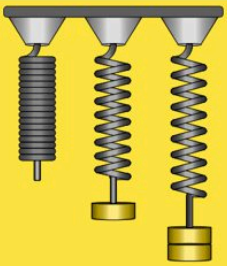
Oscillations

Definition: Oscillations are repetitive back-and-forth movements about a central point.
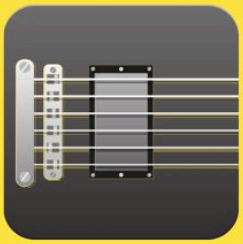
Examples:



A pendulum swinging



A mass bouncing on a spring



A guitar string vibrating



A seesaw moving up and down

18.1 Types of Oscillations

Oscillations in Real Life



Timekeeping: Clocks use pendulums or quartz oscillations to keep time.



Musical Instruments: Sound is produced through vibrating strings, air columns, or membranes.



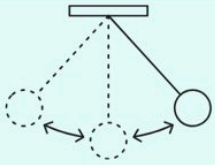
Communication: Radio waves, Wi-Fi, and mobile signals rely on electromagnetic oscillations.



Seismology: Studying earthquake waves to understand the Earth's movements.

18.1 Types of Oscillations

Types of Oscillations

	Definition	Illustration
Free Oscillations	Object oscillates on its own after being displaced, with no continuous external force. ***Eg. A pendulum swinging after being pushed.***	
Forced Oscillations	Object is kept oscillating by a continuous external force. ***Eg. A child on a swing being pushed repeatedly.***	

18.2 Oscillations in Action

Visualizing Oscillations

Here are some ways to observe oscillations when the frequency is too high for the human eye to detect:



A stroboscope flashes light at specific intervals to make fast-moving objects appear as if they are moving slowly or even stationary.



A high-speed camera can capture oscillations at an extremely high frame rate and then play them back in slow motion.

18.3 Characteristics of Oscillatory Motion

Motion in Oscillations

- An oscillating object moves back and forth around an equilibrium position.
- Characteristics of the movement:



Maximum Speed: When passing through the center (equilibrium position), the object moves at its fastest speed.



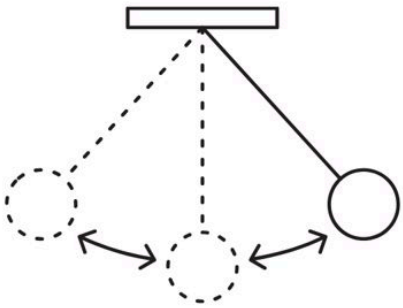
Slowing Down: As the object moves toward its extreme positions, its speed decreases.



Momentary Stop & Direction Change: At the extreme positions, the object comes to a brief stop before reversing direction.

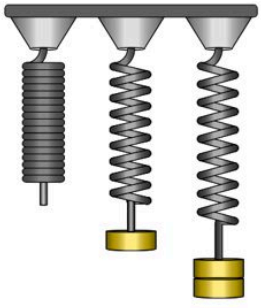
18.3 Characteristics of Oscillatory Motion

Examples



A pendulum:

- Swings **fastest** at the lowest point
- **Slows down** as it reaches the highest points.



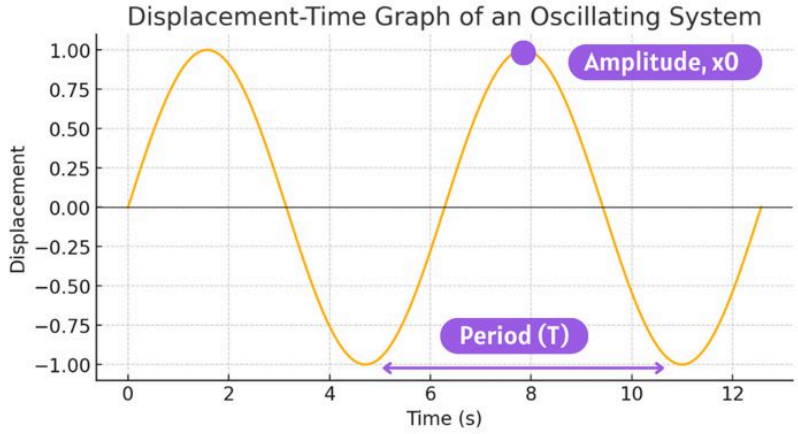
A spring-mass system:

- **Moves fastest** when passing the equilibrium point
- **Slows** at maximum compression/stretch

18.3 Characteristics of Oscillatory Motion

Displacement-Time Graph of an Oscillator

- The displacement-time graph of an oscillating system follows a **sinusoidal wave.**
- **Key Features:**



Important Formula

$$frequency = \frac{1}{period} = f = \frac{1}{T}$$

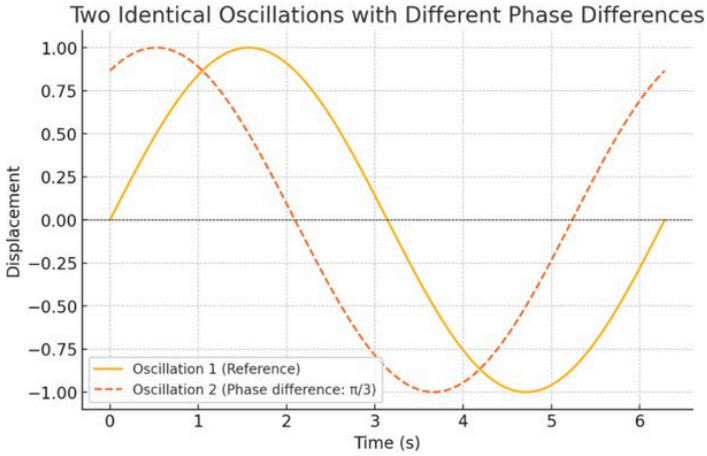
$$period = \frac{1}{frequency} = T = \frac{1}{f}$$

18.3 Characteristics of Oscillatory Motion

Phase Difference

Phase - The position of a point in an oscillation cycle at a given time. It tells us where the oscillating object is within its repetitive motion.

Phase Difference - How much one oscillation is ahead or behind another. It is the angular difference between two oscillating systems.

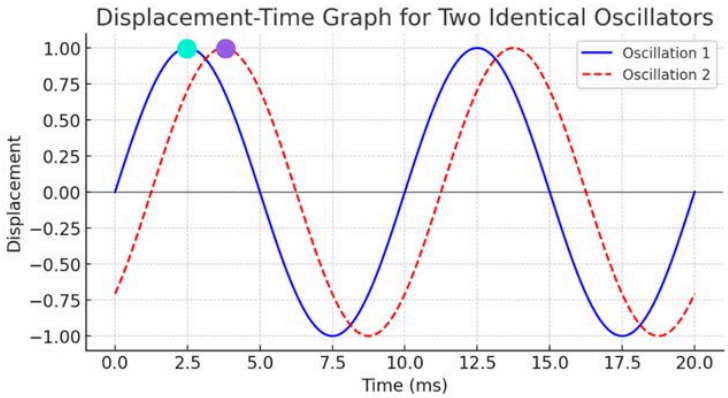


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18.3 Characteristics of Oscillatory Motion

Worked Example:

Calculate the phase difference (in degree) between the following two oscillations.



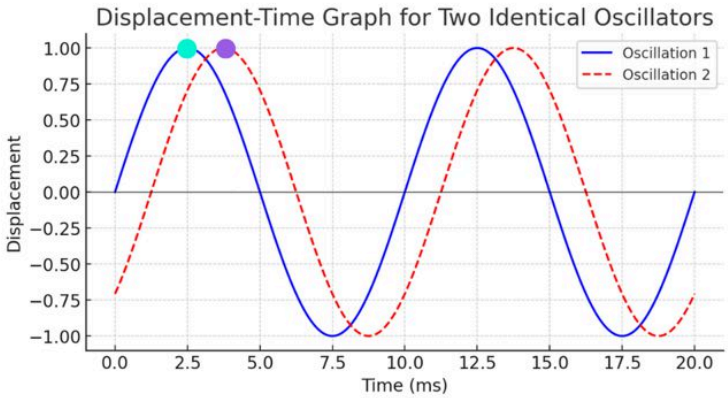
Step I: Determining the period

$$T = 10ms$$

18.3 Characteristics of Oscillatory Motion

Worked Example:

Calculate the phase difference (in degree) between the following two oscillations.



Step I: Determining the period

$$T = 10ms$$

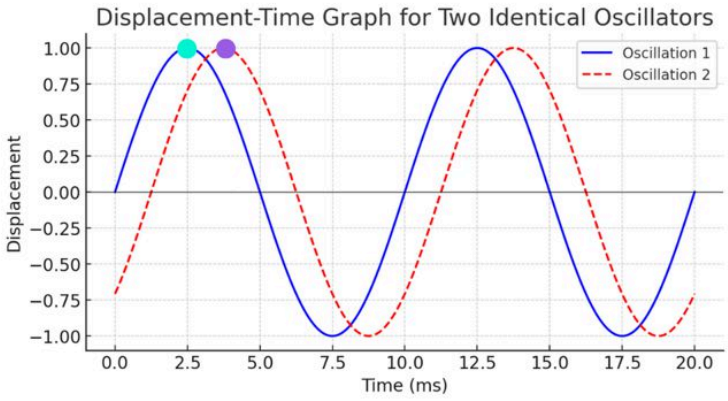
Step 2: Measure the time intervals between two points

$$\Delta t = 1.25\ ms$$

18.3 Characteristics of Oscillatory Motion

Worked Example:

Calculate the phase difference (in degree) between the following two oscillations.



Step I: Determining the period

$$T = 10ms$$

Step 2: Measure the time intervals between two points

$$\Delta t = 1.25\ ms$$

Step 3: Find the phase difference

$$\frac{\Delta t}{T} \times 360° = \frac{1.25}{10} \times 360° = 45°$$

18.4 Fundamentals of Simple Harmonic Motion

Simple Harmonic Motion

SHM is a **special type of oscillation** where the
restoring force is always directed toward the equilibrium position
and is **proportional to displacement**.

18.4 Fundamentals of Simple Harmonic Motion

Simple Harmonic Motion

Oscillation just means moving back and forth repeatedly, like a swing or a pendulum. In Simple Harmonic Motion (SHM), this movement is smooth and follows a regular pattern (like a sine wave).



SHM is a **special type of oscillation** where the **restoring force is always directed toward the equilibrium position** and is **proportional to displacement**.

18.4 Fundamentals of Simple Harmonic Motion

Simple Harmonic Motion

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SHM is a **special type of oscillation** where the **restoring force is always directed toward the equilibrium position** and is **proportional to displacement**.

This is the force that pulls or pushes the object back to its resting position (called the equilibrium position).

18.4 Fundamentals of Simple Harmonic Motion

Simple Harmonic Motion

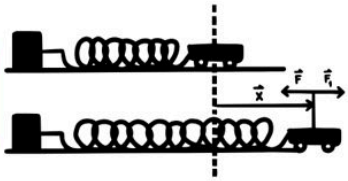
Oscillation just means moving back and forth repeatedly, like a swing or a pendulum. In Simple Harmonic Motion (SHM), this movement is smooth and follows a regular pattern (like a sine wave).



SHM is a **special type of oscillation** where the **restoring force is always directed toward the equilibrium position** and is **proportional to displacement**.

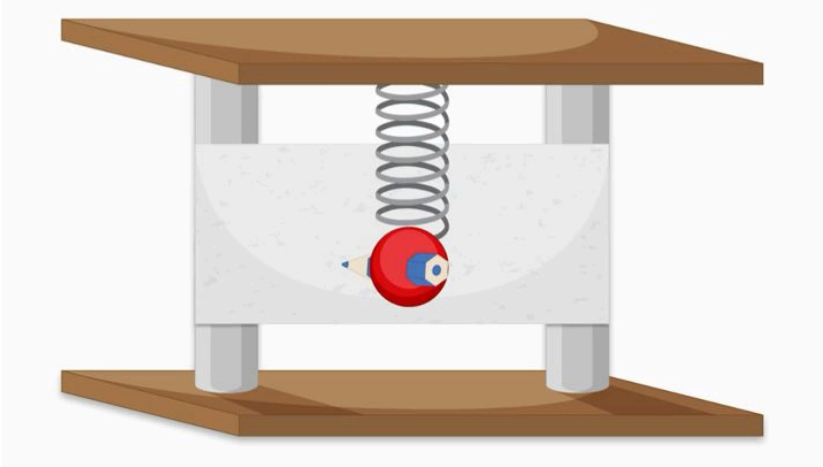
This is the force that pulls or pushes the object back to its resting position (called the equilibrium position).

In SHM, the restoring force gets stronger the further you move from the center.



18.4 Fundamentals of Simple Harmonic Motion

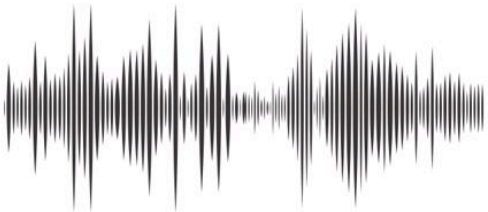
Simple Harmonic Motion



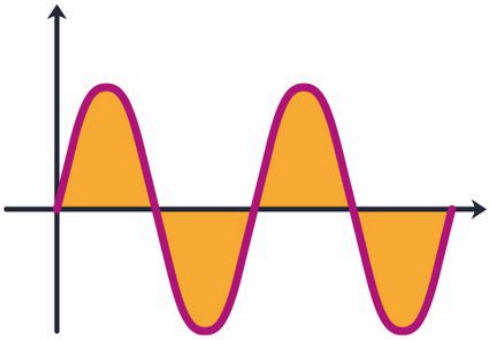
The motion is sinusoidal, meaning it follows a smooth, wave-like pattern over time.

18.4 Fundamentals of Simple Harmonic Motion

Examples



Sound Waves: Air molecules vibrate back and forth in SHM, creating pressure waves that we hear as sound.



Alternating Current (AC): In an electrical circuit, the voltage and current oscillate sinusoidally, behaving similarly to SHM.

18.4 Fundamentals of Simple Harmonic Motion

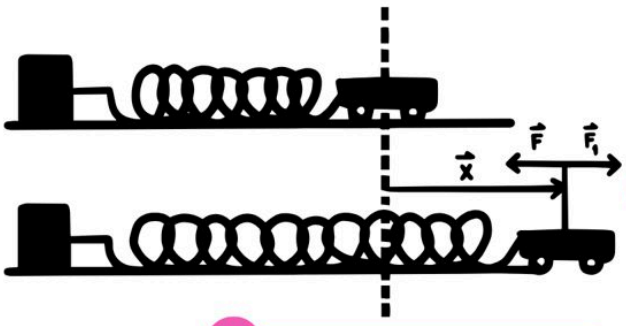
Complex Oscillations & Frequency Breakdown

- Not all oscillations are simple. A real vibrating object (e.g., a drum) can have multiple frequencies at once.
- Mathematically, any complex oscillation can be broken into simpler oscillations of different frequencies (Fourier analysis).
- This idea is important in sound processing, signal analysis, and engineering.



18.4 Fundamentals of Simple Harmonic Motion

Three Requirements for SHM in a Mechanical System



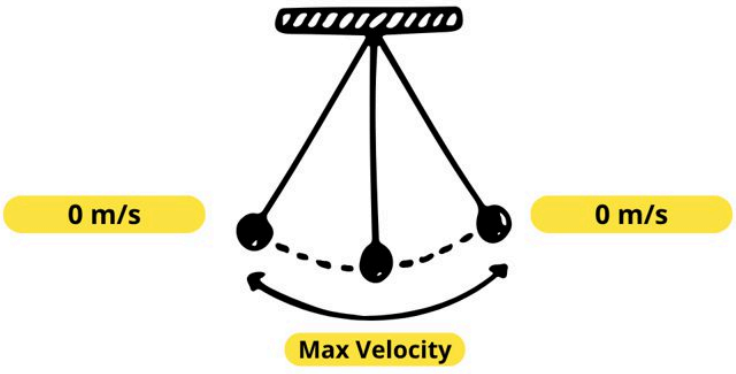
1 ***A mass that oscillates:*** Something that can move back and forth.

2 ***An equilibrium position:*** The point where the object naturally rests if undisturbed.

3 ***A restoring force that brings it back:*** A force must always act to return the object toward equilibrium.

18.4 Fundamentals of Simple Harmonic Motion

Changes in Velocity in SHM (Example: Pendulum Swing)



- **Velocity is maximum** when the object passes through equilibrium.
- **Velocity is zero** at the extreme positions (when the object momentarily stops before reversing direction).
- **Acceleration is greatest at the extreme positions** because the restoring force is strongest there.

18.5 Visualizing Simple Harmonic Motion

Experiment to Demonstrate SHM Graphically

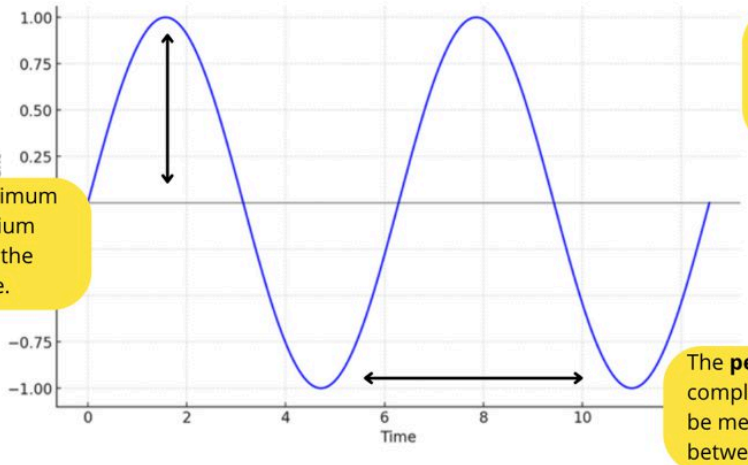


1. Attach the mass to the spring and allow it to hang at its **equilibrium position**.
2. **Pull** the mass down slightly and release it, allowing it to oscillate vertically.
3. Position the **motion sensor** below the mass to record its movement over time.
4. Use computer software to generate real-time graphs of displacement, velocity, and acceleration as functions of time.

18.5 Visualizing Simple Harmonic Motion

Displacement-Time Graph

This graph shows how the position of the mass changes over time.



It has a **sinusoidal shape**, indicating periodic motion back and forth.

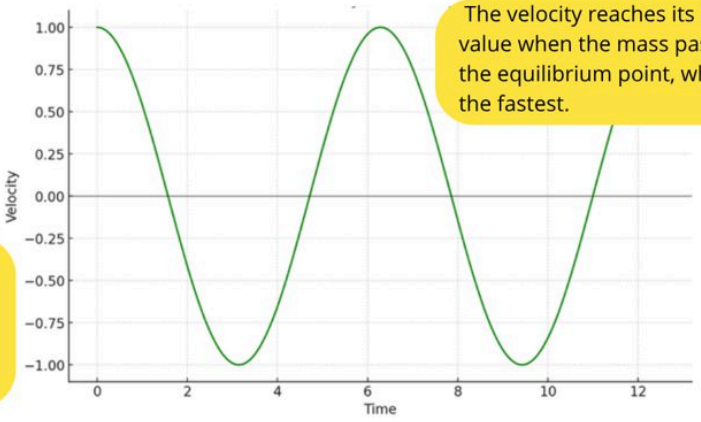
The **amplitude** is the maximum distance from the equilibrium position, showing how far the mass moves on either side.

The **period** is the time taken for one complete cycle of motion, which can be measured by finding the distance between two peaks or troughs.

18.5 Visualizing Simple Harmonic Motion

Velocity-Time Graph

The velocity graph is also sinusoidal but starts at a different point compared to displacement.



The velocity reaches its **maximum** value when the mass passes through the equilibrium point, where it moves the fastest.

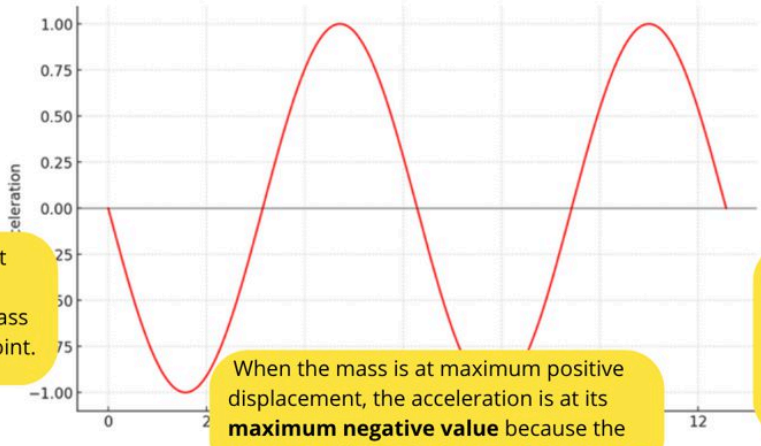
It is **90 degrees out of phase** with the displacement graph. When displacement is maximum (at the peak), velocity is zero because the mass changes direction.

You can find the velocity graph by calculating the **gradient (slope)** of the displacement-time graph, as velocity is the rate of change of displacement.

18.5 Visualizing Simple Harmonic Motion

Acceleration-Time Graph

The acceleration graph is also sinusoidal but is always in the opposite direction of the displacement graph.



It has **negative acceleration** at the beginning because the restoring force is pulling the mass back toward the equilibrium point.

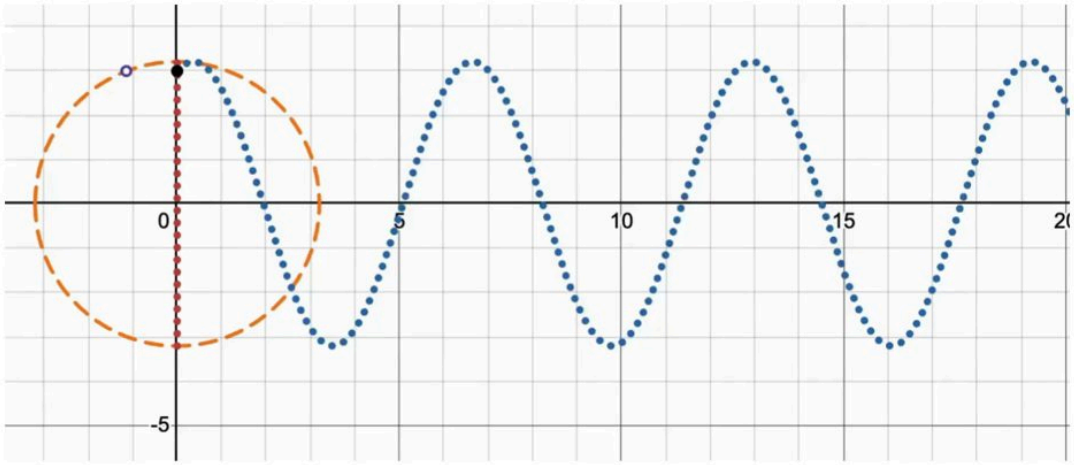
When the mass is at maximum positive displacement, the acceleration is at its **maximum negative value** because the spring is pulling it back toward the center.

The shape of this graph helps to understand how the **restoring force** works continuously to pull the mass back to the equilibrium position.

18.6 Angular Frequency

What is Angular Frequency?

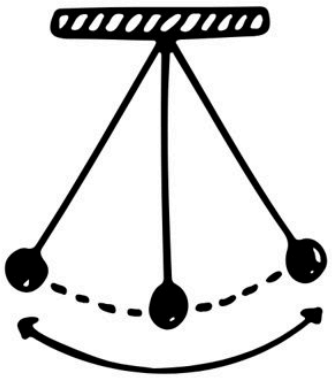
- We know frequency (f) is how many oscillations happen in one second (measured in Hz).
- But in circular motion and oscillations, we often use angular frequency instead.



18.6 Angular Frequency

Why Do We Use Angular Frequency?

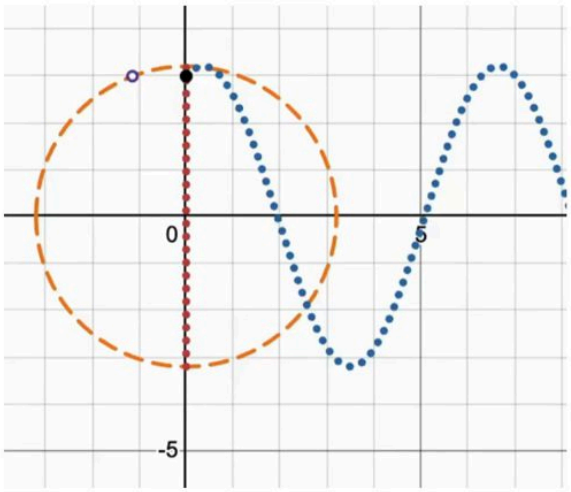
- Frequency tells us how many complete oscillations per second.
- But oscillations are often **connected to circular motion** (e.g., pendulums, springs, waves).
- Angular frequency helps describe motion in terms of angles (radians).



18.6 Angular Frequency

What is Angular Frequency?

Definition: **Angular frequency** tells us how many radians an oscillation sweeps through per second.



Formula

$$\omega = 2\pi f$$

Since $f = \frac{1}{T}$,

Formula

$$\omega = \frac{2\pi}{T} \quad \text{or} \quad T = \frac{2\pi}{\omega}$$

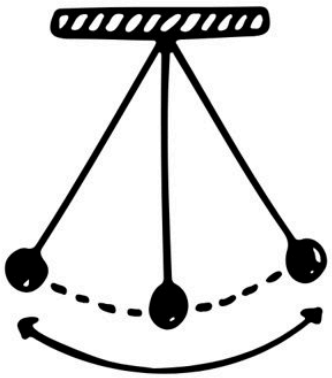
18.6 Angular Frequency

Why Do We Use Angular Frequency?

- Frequency tells us how many complete oscillations per second.
- But oscillations are often **connected to circular motion** (e.g., pendulums, springs, waves).
- Angular frequency helps describe motion in terms of angles (radians).

A pendulum swinging back and forth 2 times per second has:

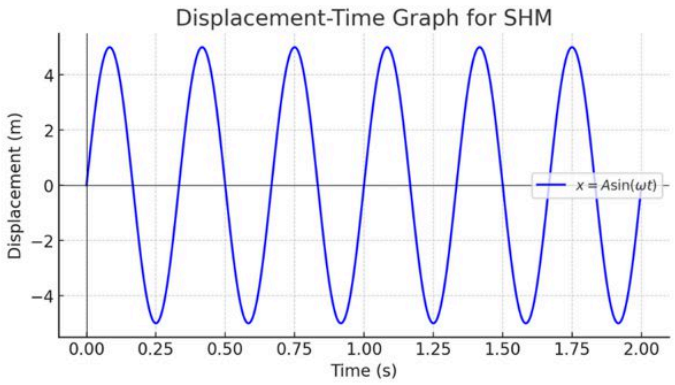
$\omega = 2\pi(2) = 4\pi$ rad / s



18.6 Angular Frequency

Worked Example

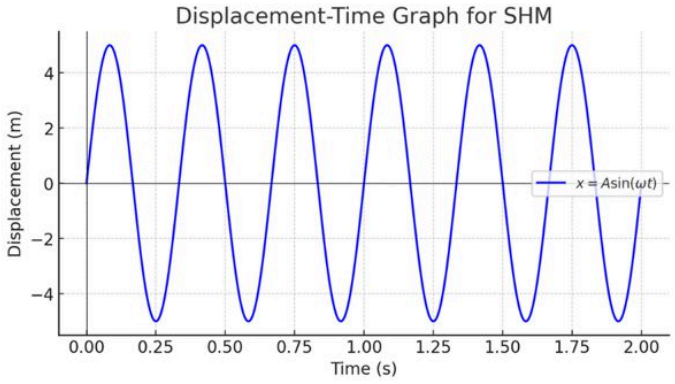
What is the angular frequency of the following displacement-time graph?



18.6 Angular Frequency

Worked Example

What is the angular frequency of the following displacement-time graph?



$$\omega = 2\pi(3) = 6\pi$$

18.7 More Equations

Equations of Harmonic Motion

Simple harmonic motion (SHM) follows a sinusoidal pattern, meaning its displacement can be described using either the sine or cosine function:

Formula

$$x = x_0 \sin \omega t$$

Formula

$$x = x_0 \cos \omega t$$

x_0 (amplitude) – The maximum displacement from the equilibrium position.

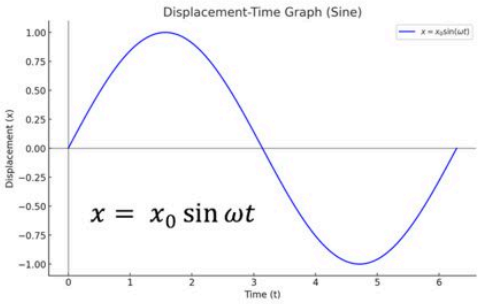
ω (angular frequency) – How fast the oscillation completes one cycle.

t (time) – The time elapsed since the motion started.

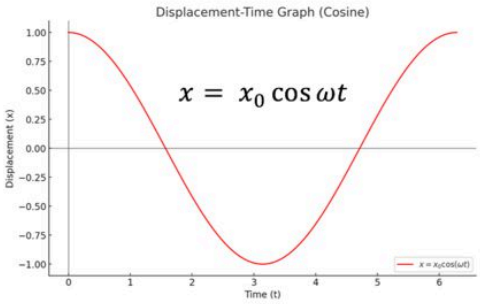
18.7 More Equations

Understanding the Difference

The two equations are similar but describe motion that starts from different points:



✅ The object starts from the equilibrium position (zero displacement) at t = 0 .
✅ Example: A pendulum released from the center moving outward.



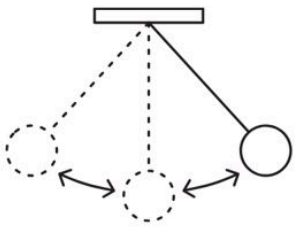
✅ The object starts from maximum displacement at t = 0 .
✅ Example: A pendulum pulled to one side and released.

18.7 More Equations

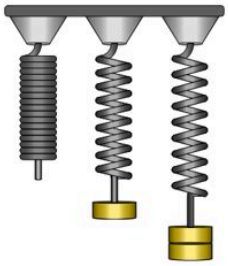
Why Do We Need These Equations?

Oscillations in SHM repeat over time, so we need a mathematical way to predict the position of an object at any given moment.

Examples of SHM systems that follow these equations:



A pendulum swinging



A mass bouncing on a spring

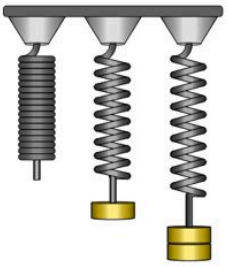
many more!

18.7 More Equations

Worked Example

A mass-spring system oscillates with an amplitude of 0.2 m and an angular frequency of π rad/s. At t = 0 , the mass is at its maximum displacement.

1. Write the equation of motion for this system.
2. Find the displacement at t = 1.5 seconds.



Q1

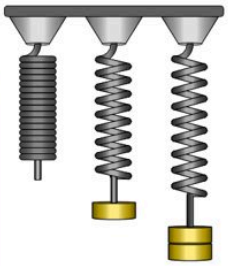
Q2

18.7 More Equations

Worked Example

A mass-spring system oscillates with an amplitude of 0.2 m and an angular frequency of π rad/s. At t = 0 , the mass is at its maximum displacement.

1. Write the equation of motion for this system.
2. Find the displacement at t = 1.5 seconds.



Q1

Since the mass is at its maximum displacement at t = 0 , we use:

$$x = x_0 \cos \omega t$$

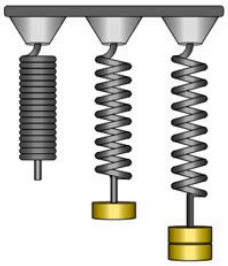
Q2

18.7 More Equations

Worked Example

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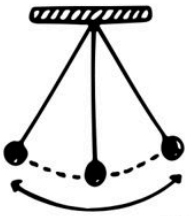
$$x = x_0 \cos \omega t$$

Q2

$$x = x_0 \cos \omega t = 0.2 \cos(\pi \times 1.5) = 0m$$

18.7 More Equations

Acceleration in Simple Harmonic Motion (SHM)



Equation

$$a = -\omega^2 x$$

Why Acceleration is Opposite to Displacement?

The negative sign means acceleration always points towards equilibrium.

Why Angular Frequency Appears in the Equation?

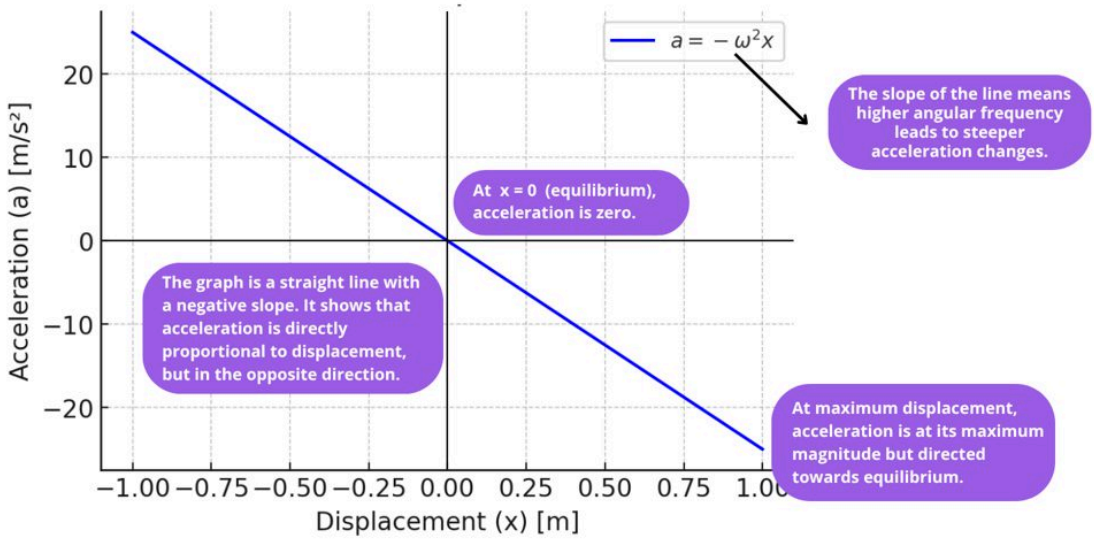
Since acceleration is the rate of change of velocity, a higher angular frequency means a greater acceleration for a given displacement.

Acceleration is directly proportional to displacement.

Eg. This means the farther the pendulum is displaced, the stronger the restoring force, and the greater the acceleration.

18.7 More Equations

Explanation of the Acceleration-Displacement Graph



18.7 More Equations

Equations for velocity

Equation that shows that v depends on x (oscillator's displacement)

$$v = \pm\omega\sqrt{x_0{}^2 - x^2}$$

Maximum speed; x = 0

Formula for maximum speed

$$v = \omega x_0$$

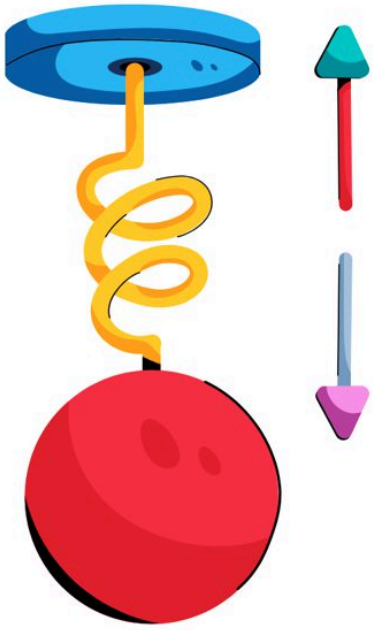
18.7 More Equations

Equations for velocity

$$v = \omega x_0$$

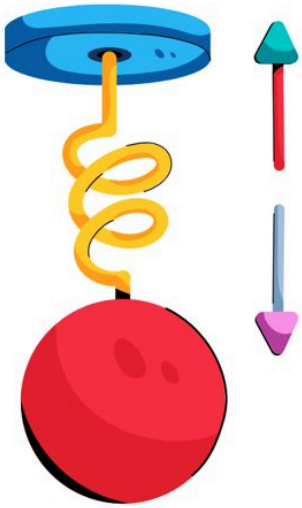
★ Greater amplitude = Greater Maximum Speed
★ The object travels a longer distance in the same time, so it must move faster

★ Greater frequency = Greater Maximum Speed
★ Higher frequency, faster oscillations, greater speed.



18.7 More Equations

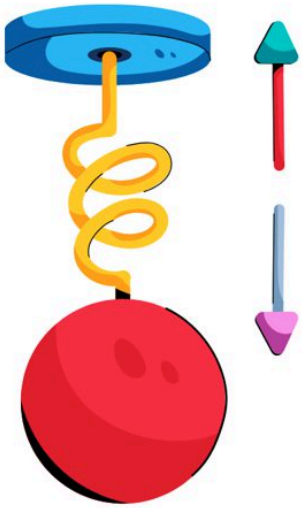
Worked Example



A mass on a spring oscillates with an amplitude of 0.2 m and a frequency of 5 Hz. Find the maximum speed of the mass.

18.7 More Equations

Worked Example



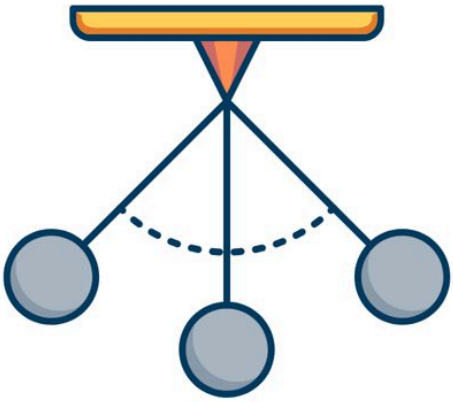
A mass on a spring oscillates with an amplitude of 0.2 m and a frequency of 5 Hz. Find the maximum speed of the mass.

$$v = \omega x_0 = (2\pi 5)(0.2) = 2\pi = 6.28\, m/s$$

18.8 Conversion of energy in SHM

Introduction to Energy in SHM

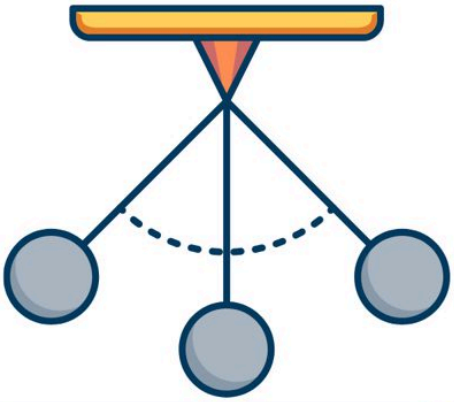
- In SHM, energy continuously transfers between potential energy (PE) and kinetic energy (KE).
- The total mechanical energy remains constant if the system is undamped (no external forces like friction).



18.8 Conversion of energy in SHM

SHM Example: Pendulum

Energy transformation occurs as follows:



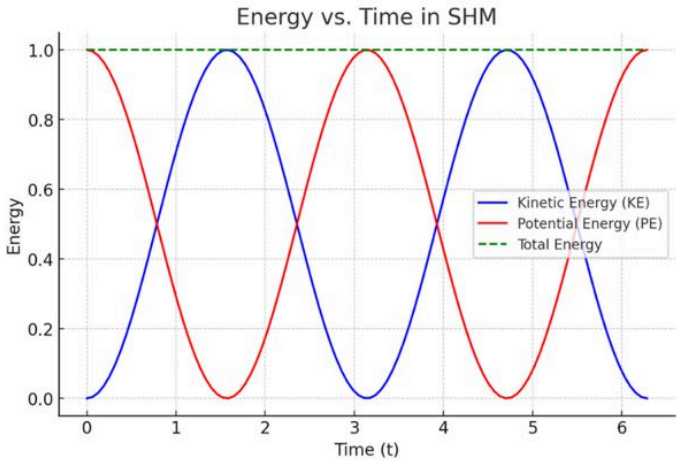
⬆️ **Highest point:**
- **PE is maximum (GPE is highest)**
- **KE is zero (no movement)**

➡️ **Between Maximum Displacement and Equilibrium**
- **As the pendulum moves down, PE decreases while KE increases.**
- **As it moves up, KE decreases while PE increases.**

⬇️ **At Equilibrium Position (x = 0, Lowest Point)**
- **KE is maximum (pendulum moves fastest).**
- **PE is zero (no height difference).**

18.8 Conversion of energy in SHM

Energy Time Graph



Potential Energy (PE) Line

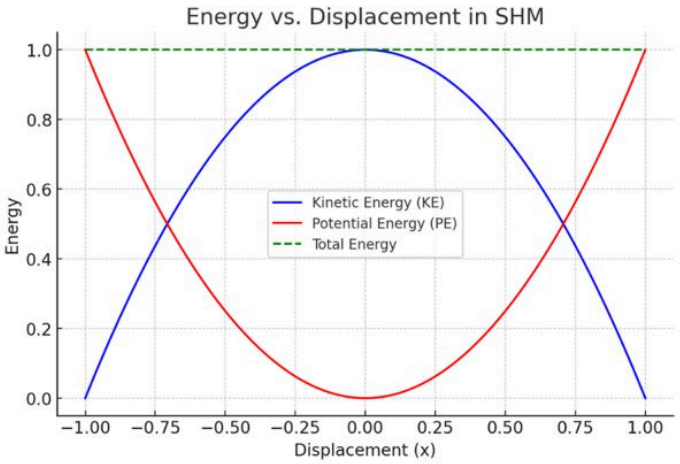
- Starts at a maximum.
- Decreases to zero at equilibrium.
- Increases again at the other extreme.

Kinetic Energy (PE) Line

- Starts at zero.
- Increases to a maximum at equilibrium.
- Decreases back to zero at maximum displacement.

18.8 Conversion of energy in SHM

Energy Displacement Graph



Potential Energy (PE) Line

- Starts at a maximum.
- Decreases to zero at equilibrium.
- Increases again at the other extreme.

18.8 Conversion of energy in SHM

Equations for Energy

Formula of Kinetic Energy

$$E_k = \frac{1}{2} mv^2$$

Formula of Maximum Velocity

$$v = \omega x_0$$

18.8 Conversion of energy in SHM

Equations for Energy

Formula of Kinetic Energy

$$E_k = \frac{1}{2} mv^2$$

Formula of Maximum Velocity

$$v = \omega x_0$$

The total energy in SHM can be calculated using the kinetic energy at the equilibrium position since, at that point, the potential energy is zero.

18.8 Conversion of energy in SHM

Equations for Energy

Formula of Kinetic Energy

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Formula of Maximum Velocity

$$v = \omega x_0$$

The total energy in SHM can be calculated using the kinetic energy at the equilibrium position since, at that point, the potential energy is zero.

$$E_k = \frac{1}{2} m(\omega x_0)^2$$

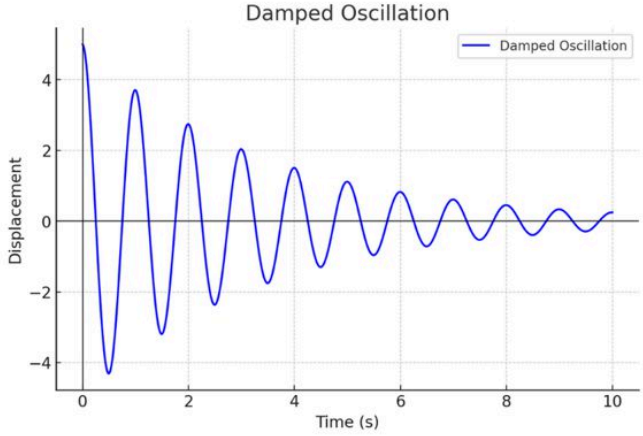
$$E_k = \frac{1}{2} m\omega^2 {x_0}^2$$

18.9 Damped Oscillations

What is Damped Oscillations?

Definition: Damped oscillations occur when the amplitude of an oscillating system gradually decreases over time due to energy loss.

Causes: Energy is dissipated due to friction, air resistance, or internal forces in the material.



18.9 Damped Oscillations

Why Does Amplitude Decrease Exponentially?

Exponential Decay: The amplitude of a damped oscillation decreases exponentially over time, meaning it reduces by a fixed proportion in each cycle.
Why? As the object slows down, the resistive forces acting on it (e.g., air resistance) also decrease, leading to a gradual reduction in amplitude rather than a sudden stop.



18.9 Damped Oscillations

Relationship Between Amplitude and Frequency

Key SHM Property: In simple harmonic motion, frequency depends only on system properties (like mass and stiffness) and is independent of amplitude.
Implication: Even as damping reduces amplitude, the frequency of oscillation remains nearly unchanged unless heavy damping occurs.



18.9 Damped Oscillations

Washing machine drum

A washing machine drum during the spin cycle is an example of **useful damping.**

When the drum spins rapidly, any imbalance in the clothes can cause vibrations. To prevent excessive shaking, washing machines have **dampers and shock absorbers** that gradually reduce the oscillations.



18.10 Resonance

Resonance

Definition

Resonance occurs when an object vibrates with **maximum amplitude** due to a matching external force at its **natural frequency.**

What is Natural Frequency?



A tuning fork produces sound at its natural frequency when struck.



A bridge sways at a certain frequency when the wind blows.

18.10 Resonance

Real-Life Example of Resonance

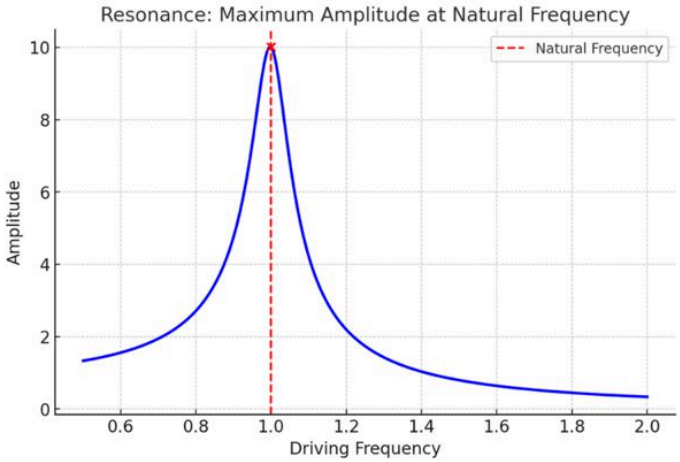
Imagine a singer breaking a glass with their voice.

- Every glass has a **natural frequency** at which it vibrates.
- If the singer produces a sound **matching** this frequency, the glass starts vibrating.
- With enough time and the right volume, the vibrations build up, causing the glass to shatter due to excessive amplitude.



18.10 Resonance

Resonance and Amplitude Growth

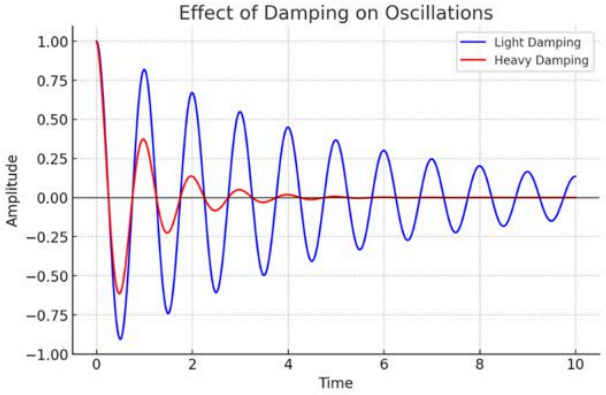


- When the forcing frequency (external driving force) matches an object's natural frequency, its amplitude increases significantly.
- This explains why bridges and buildings can collapse if external forces (wind, earthquakes) match their natural frequency.

18.10 Resonance

The Role of Damping in Resonance

- Damping reduces amplitude in resonance.
- If there is light damping, the oscillations grow large before gradually reducing.
- If there is heavy damping, resonance is minimized, preventing excessive oscillations.



18.10 Resonance

Critical Damping

Definition

Critical damping occurs when the damping force is **just enough** to bring an oscillating system back to equilibrium in the shortest possible time without overshooting or oscillating.

Car Suspension System

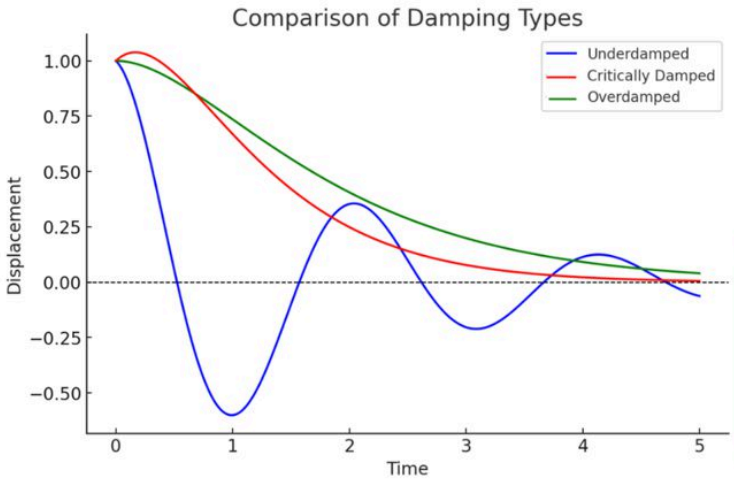


Shock absorbers help a car stabilize quickly after a speed bump. Underdamping causes bouncing, overdamping slows recovery, while critical damping ensures a smooth, fast return.

18.10 Resonance

Critical Damping Graph

Underdamped (blue): Oscillates before settling at zero.



Overdamped (green): Always above the critically damped curve, showing it takes longer to reach equilibrium.

Critically damped (red): Returns to zero quickly without oscillating.

18.10 Resonance

Useful Applications of Resonance



Microwave Ovens: Water molecules resonate with microwave radiation, heating food efficiently.



Musical instruments: Strings and air columns resonate to amplify sound.



MRI machines: Resonance is used in medical imaging to detect different tissues in the body.



Radio tuning: A radio selects the correct station by adjusting resonance with the desired frequency.